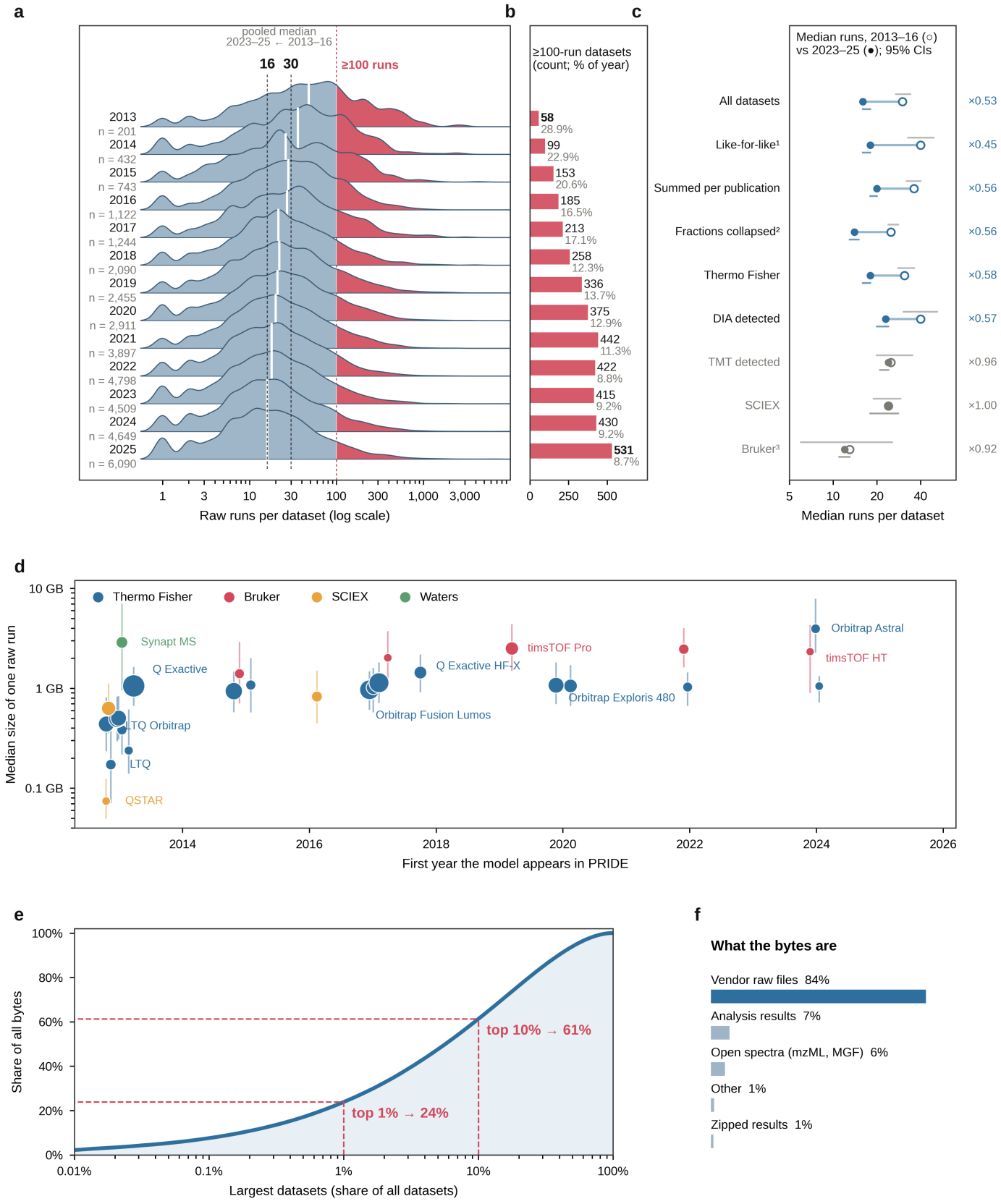

a
pooled median
2023–25 ← 2013–16
16
30
≥100 runs
2013
n = 201
2014
n = 432
2015
n = 743
2016
n = 1,122
2017
n = 1,244
2018
n = 2,090
2019
n = 2,455
2020
n = 2,911
2021
n = 3,897
2022
n = 4,798
2023
n = 4,509
2024
n = 4,649
2025
n = 6,090
1
3
10
30
100
300
1,000
3,000
Raw runs per dataset (log scale)
b
≥100-run datasets
(count; % of year)
58
28.9%
99
22.9%
153
20.6%
185
16.5%
213
17.1%
258
12.3%
336
13.7%
375
12.9%
442
11.3%
422
8.8%
415
9.2%
430
9.2%
531
8.7%
0
250
500
c
Median runs, 2013–16 (○)
vs 2023–25 (●); 95% CIs
All datasets
×0.53
Like-for-like¹
×0.45
Summed per publication
×0.56
Fractions collapsed²
×0.56
Thermo Fisher
×0.58
DIA detected
×0.57
TMT detected
×0.96
SCIEX
×1.00
Bruker³
×0.92
5
10
20
40
Median runs per dataset
d
Thermo Fisher
Bruker
SCIEX
Waters
10 GB
1 GB
0.1 GB
Median size of one raw run
Synapt MS
Q Exactive
LTQ Orbitrap
LTQ
QSTAR
Q Exactive HF-X
Orbitrap Fusion Lumos
timsTOF Pro
Orbitrap Exploris 480
Orbitrap Astral
timsTOF HT
2014
2016
2018
2020
2022
2024
2026
First year the model appears in PRIDE
e
100%
80%
60%
40%
20%
0%
Share of all bytes
top 10% → 61%
top 1% → 24%
0.01%
0.1%
1%
10%
100%
Largest datasets (share of all datasets)
f
What the bytes are
Vendor raw files 84%
Analysis results 7%
Open spectra (mzML, MGF) 6%
Other 1%
Zipped results 1%